

Short Read Workshop Day 4

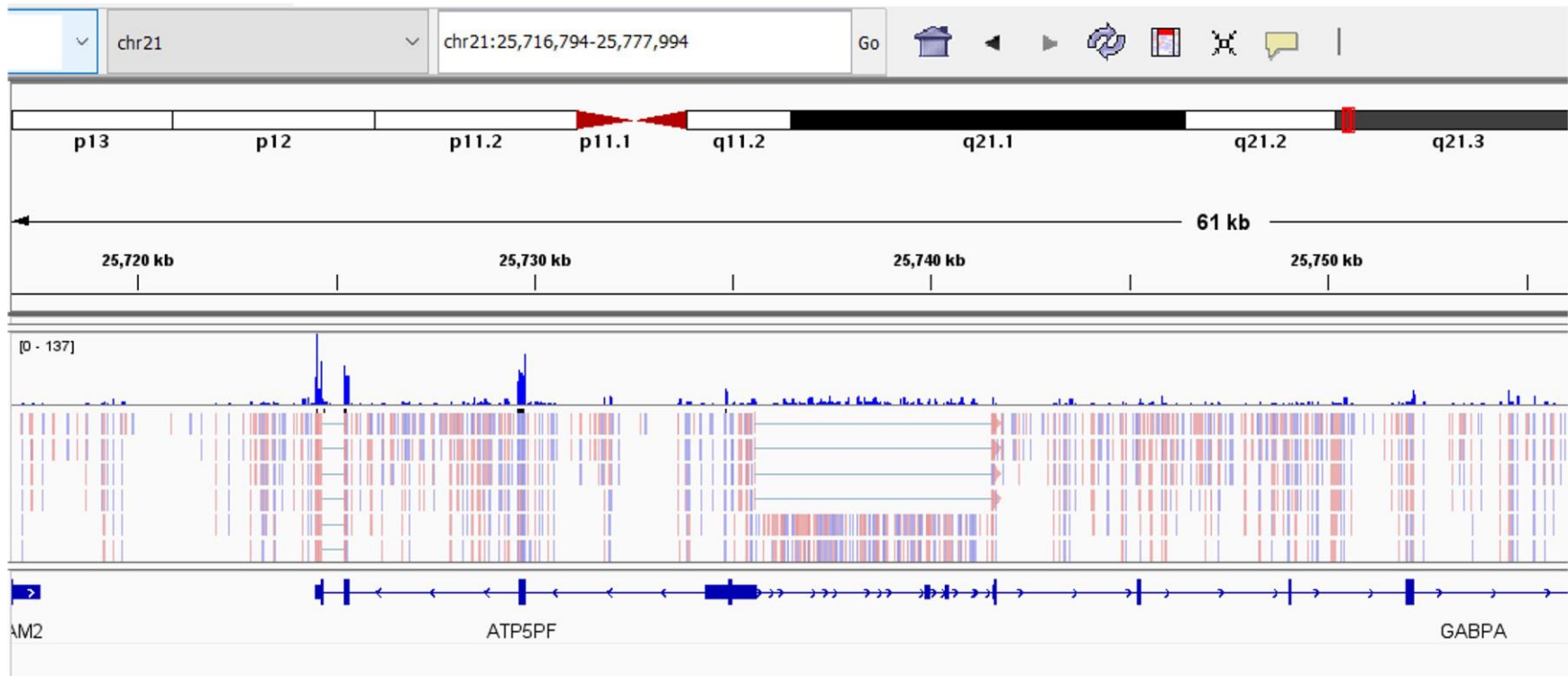
Trimming, Mapping, IGV

Day 4 overview

- Trimming fastq files
- Mapping fastq files
- More about mapped file formats
- Visualizing mapped files

Goal of the Day

View sequencing data as reads aligned to a genome



Trimming/mapping recap

50 base read:

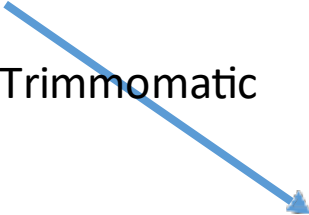
TAGGCTAACTCTGTAGCCCCAGGTACCATGCATAATTGAC**CAGGATATAG**

Trimming/mapping recap

50 base read:

TAGGCTAACTCTGTAGCCCCAGGTACCATGCATAATTGAC**CAGGATATAG**

Trimmomatic



40 base trimmed read:

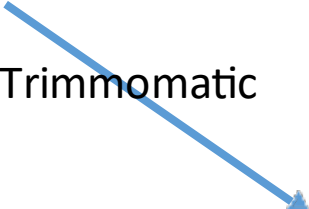
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HISAT2



Genome:

AGCTTCGGATCGATCGACTGAC**TAGGCTAACTCTGTAGCCCCAGGTACCATGCATAATTGAC**CGCGATTACGAC
TCGAAGCCTAGCTAGCTGACTGATCCGATTGAGACATCGGGGTCCATGGTACGTATTAAGTGGCGCTAATGCTG

Trimming/mapping recap

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Trimmomatic

40 base trimmed read:

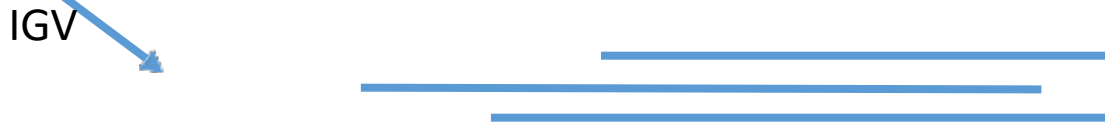
TAGGCTAACTCTGTAGCCCCAGGTACCATGCATAATTGAC

HISAT2

Genome:

AGCTTCGGATCGATCGACTGACT**TAGGCTAACTCTGTAGCCCCAGGTACCATGCATAATTGAC**CGCGATTACGAC
TCGAAGCCTAGCTAGCTGACTGATCCGATTGAGACATCGGGGTCCATGGTACGTATTA**ACTGGCGCTAATGCTG**

IGV



AGCTTCGGATCGATCGACTGACTTAGGCTAACTCTGTAGCCCCAGGTACCATGCATAATTGACCGCGATTACGAC
TCGAAGCCTAGCTAGCTGACTGATCCGATTGAGACATCGGGGTCCATGGTACGTATTA**ACTGGCGCTAATGCTG**

Trimming fastq files with Trimmomatic

- Follow Variables/Trimmomatic worksheets to:
- Learn more about variables in bash
- Create Day4 directories
- Edit script to run Trimmomatic
 - Input: fastq files with full-length reads
 - Output: fastq files with trimmed/filtered reads
- Extra: Edit the d4_trim_qc.sbatch script to run pre-trim qc and compare the plots/stats to post-trim

How do you trim polyA regions from
both sides of reads?

How do you trim polyA regions from both sides of reads?

- Make a new fasta file with a polyA segment, or append to the Illumina adapter file, if writeable

- >polyA

AAAAAAAAAAAAAAAAAAAAAAAAAAAA

- ILLUMINACLIP:<new fasta file>:2:30:10

Mapping fastq files with HISAT2

- Follow Mapping/IGV worksheet to:
- Rsync the mapping script
- Edit script and run HISAT2

Input: trimmed fastq files

Outputs: .sam, .bam, .sorted.bam, .bam.bai files

- Visualize BAM file on the IGV web app

Homework

Day 4 Homework – FASTQC, trimming, mapping, IGV

The assessment tomorrow will run many of the same steps as this homework. These steps are essential in ALL short read data processing.

Variables – evaluating (calling)

\$variablename

echo \$a

grep \$gene_name <filename>

trim_script="\$filepath"/d4_trim_qc.sbatch

OR

trim_script=\${filepath}/d4_trim_qc.sbatch

wc \$filelist

Several ways of evaluating:

\$a

\${a}

“\$a”

These differ slightly, and you will see us use them all in scripts